Lumbini Technological University

Institute of Engineering and IT

Banke, Lumbini Province, Nepal

B. Tech. in CS and AI

Course title- **Physics for Computing**	Year/Part: I/I	Credit 3 (48 hrs)
Course code	Semester: I	Lecture – 2 Tutorial 0 Practical 1

1. Introduction

This Course has been designed keeping in view with recent development in computational physics. It combines computer science, physics and applied mathematics to develop scientific solutions of complex problems. It complements the area of theory and experiments in scientific investigations. It can be applied in molecular modelling, electronic circuit design, protein folding, atmospheric science, aerodynamic design and testing and material science. Availability of additional specialties in areas and upgraded quality of course will relate physics with CS technology. Various topics are incorporated in the syllabus to relate physics and computer science along with information technology.

2. Objectives

The course is designed with following objectives:

- To provide students up to date knowledge of fundamentals of computing science.
- To impart knowledge of basic view of physical principles of theoretical and experimental aspects of computers.
- To acquaint students with the recent trends in physics and electronics.
- To develop manpower in computer engineering at the tertiary level to conduct research in engineering on the basis of physics and electronics.
- To produce high level research-oriented man power in physics so that they can be able to tackle high tech IT industry.

3. Learning Outcome

After the completion of the course, the student will be able to

- Understand fundamental concepts of physics principle underlying computers.
- Develop knowledge about performance, power and thermal behavior of computers.
- Interrelate physics theoretically and practically with computer.
- Know to operate instruments of related field.

4. Course contents (2 credits/32 hrs)

1. Semiconductor and semiconductor devices (4 hrs)

Intrinsic and extrinsic semiconductors, Electrical conductivity of semiconductors, photoconductivity, metal-metal junction, contact potential, semiconductor diode, bipolar junction transistor (BJT), Field effect transistor

2. Universal gates and physics of integrated circuits (7 hrs)

Universal gates, RTL and TTL gates, clock circuits, semiconductor purification, Zone refining, single crystal growth, process of IC production, electronic component fabrication on a chip, CMOS inverter, static gate characteristics, delay, power and energy, scaling theory, reliability.

3. Sequential machines (3 hrs)

Introduction, combinational logic, interconnect, sequential machines

4. Processors and systems (6 hrs)

Introduction, system reliability, processors, memory, mass storage, system power consumption, heat transfer

5.Input and output (4 hrs)

Introduction, displays, image sensors, touch sensors, microphones, accelerometers and inertial sensors

6.Emerging technologies (8hrs)

Carbon nanotubes, quantum computer, machine learning, deep learning, reinforcement learning, Robotics, natural language processing, computer vision. recommender system

Tutorial

N/A

Practical (1credit/16)

The practical work includes the fabrication of multivibrator, touch sensors, machine learning, robotics, computer vision etc.

5. Evaluation scheme

Evaluation Method	Description	Weight (%)
Assignments	Regular assignments will be given to assist the understanding and practical implementation of web technology	30
Project/Seminars	Students will work on individual and group projects to demonstrate their ability to design and develop functional web applications.	20
Class Participations	Active participations lab and workshops will be taken into account.	10
Final Examination	Final examination will be conducted to evaluate theoretical knowledge and problem solving skills.	40

6. Recommended Books

Marilyn Wolf, The Physics of Computing 1st ed.

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B. Tech. in IT Engineering/B. Tech. in CS and AI/B. Tech. in IT

Course Title: English Communication	Year/Part: I/I	Credit: 3 (48 hrs)
Course Code:	Semester: I	Lecture: 3

1. Course Description

This course *Communication Skills* has been meticulously developed to equip students with a wide range of techniques for effectively utilizing English in both academic and communicative contexts. It aims to train students in comprehending diverse texts, familiarizing them with terminologies, enhancing their grasp of grammatical nuances, and honing their communicative skills in English. By exploring the intricate relationship between language structures and meaning, this course enables students to progress from a general understanding to a more comprehensive mastery of the language.

2. Course Objectives

The course aims to achieve the following objectives:

- To acquaint the students with necessary grammar and use them to construct sensible sentences.
- To enable the students to answer the questions given after the comprehension exercises.
- To enable the students to write paragraphs on people, places, and events correctly and meaningfully.
- To familiarize the students with the various genres of literature and analyze them.
- To enable the students with the process of writing.
- To enable the students to produce different texts and genres of writing.
- To make the students capable of producing professional writing such as proposals, and report for their own project.

3. Learning Outcomes and Course Contents

Learning outcomes	Contents
Speculate the title/topic from the given texts. Find out the themes from the texts. Sketch the character. Comprehend the texts and answer the questions based on the facts and imagination.	**Unit 1: Reading (12)** 1.1 The Bet by Anton Chekov (Story) 1.2 The Gift of the Magi by O'Henry (Story) 1.3 The Road Not Taken by Robert Frost (Poem) 1.4 Lunatic by Laxmi Prasad Devkota (Poem) 1.5 Post modern mind by Richard Tarnas (Essay)

Take notes from the texts. Summarize the texts. Interpret the texts.	1.6 Humility by Noah Harari (Essay) 1.7 Alchemist by Paulo Coelho (Novel) 1.8 A Doll's House by Henrick Ibson (one-act play)
Understand the grammatical rules and their contexts/structures. Construct accurate sentences in their writing. Produce a meaningful paragraph. Use the mechanics of writing appropriately.	**Unit 2: Sentence Construction and Mechanics in Writing (6)** 2.1 Sequence of tenses 2.2 Active and passive voice 2.3 Subject-verb agreement 2.4 Conditional sentences 2.5 Prepositions 2.6 Mechanics in writing: Punctuations, spelling, capitalization, italics, abbreviations, etc.
Describe the background of writing i.e., purpose. Discuss the process of writing. Follow pre-writing activities before actual writing. Write a variety of paragraphs and essays. Write a variety of texts and genres of writing.	**Unit 3: Writing (20)** 3.1 General writing 3.1.1 Job Application and CV 3.1.2 Greeting card and invitation letter 3.1.3 Email and SMS 3.1.4 Brochures, posters, and blogs 3.1.5 Business letter 3.1.6 Notice and minute 3.1.7 Preparing memoranda 3.1.8 Essay writing 3.2 Technical writing 3.2.1 Proposal writing/project writing 3.2.2 Types of formal reports: Progress report, feasibility report, empirical report, research report, technical report 3.2.3 Report writing: Report writing: Memo report, letter report, project/field report

Use the language functions correctly and appropriately. Give the instructions/directions accurately. Describe the person, place, things, or procedures. Express their own views and ideas. Involve in group discussions. Conduct the meetings. Take and give interviews. Present their report appropriately.	**Unit 4: Oral Communication (10)** 4.1 Greeting and introducing 4.2 Giving instruction/direction 4.3 Describing 4.4 Thanking and accepting an apology 4.5 Expressing individual views 4.6 Group/round table discussions 4.7 Conduct meeting 4.8 Interview (techniques, body language, dos and don'ts) 4.9 Talk delivery 4.10 Presenting a brief report

3. **The instructional strategies will be adopted as follows.**

   Varieties of techniques/methods can be applied for the facilitation of this course. The main techniques/methods applicable to this course include lectures, teacher-guided reading, self-reading, question-answer, discussion, individual/pair/group work, project work, class presentation, role plays/simulations/dramatizations, demonstration, etc.

4. **Evaluation Scheme**

The evaluation of students' performance in the course will be based on the following components:

Evaluation Method	Description	Weight (%)
Assignments	Regular assignments will be given to assess the understanding and practical skills.	30
Projects/Seminars	Students will work on individual and group projects to demonstrate their ability in general and technical writing as well as oral communication.	20
Class Participation	Active participation in class discussions, practice, presentations and engaging in their project works/workshops will be taken into account.	10
Final examination	Final examination will be conducted to evaluate theoretical knowledge and practical skills in the English language.	40

Recommended Books:

Alred, G. J.; Brusaw, C. T. & Oliu, W.E. *The business writer's handbook* (9th edition).

Davis, F. & Rimmer, W. (2013). *Active grammar 1.* Cambridge University Press.

Lesikar, Raymond V. & et-al. *Business Communication.*

Markel, M. (2012). *Technical communication* (10th edition).

Matreyek, W. (1983). *Communicating in English.* Pergamon Press

Silyn-Roberts, H. (2013). *Writing for science and engineering* (2nd edition). Elsevier Ltd.

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B. Tech. in CS & AI

Course Title: Mathematics I **Credit: 3 (48 hrs.)**
Nature of Course: Theory

Year/Part: I/I **Lecture: 3**
Semester: I **Practical: 0**

1. Course Description

This course is designed to provide students with a comprehensive understanding basic concept and its uses to solving the problems of limits, continuity, differentiation and integration with their applications, vector algebra, analytic geometry and vector-valued functions, multivariable calculus.

2. Course Objectives

The course aims to achieve the following objectives:

- Students will be able to understand and formulate real world problems into mathematical statements.
- Students will be able to develop solutions to mathematical problems at the level appropriate to the course.
- Students will be able to describe or demonstrate mathematical solutions either numerically or graphically.

3. Methods of Instruction

Lecture, Discussion, and Class Work

4. Course Contents (3 credits/ 48 hrs.)

Unit 1: Limits and Continuity 6 hrs

Limit of a function, One-sided limits, Limits at infinity, asymptotes, Continuity of a function

Unit 2: Differentiation and its Applications 10 hrs

Derivative, Geometrical interpretation, Differentiable function at a point, Differentiability implies continuity, Derivatives of different functions, Implicit differentiation, Maximum and minimum values, Theorem (extreme value and Fermat's) (without proof), Rolle's Theorem, Mean-Value Theorem, Geometrical meaning of Mean-Value Theorem, Monotonic functions, Indeterminate forms and L'Hospital's rule, Newton's method.

Unit 3: Integration and its Applications 12 hrs

Integration, Integration by parts, Trigonometric integrals, Trigonometric substitutions, Definite integral, Properties of definite integral, Improper integrals, Area between the curves, arc length, Area of surface of revolution.

Unit 4: Vector Algebra 7 hrs

Scalar and vector product of two vectors, geometrical meaning of vector product of two vectors, Scalar and vector product of three vectors, geometrical meaning of scalar product of three vectors, reciprocal vectors of three vectors, Scalar and vector product of four vectors. Application of Vector Algebra: Projection of a vector onto another vector, Equation of line and plane by vector method

Unit 5: Analytic Geometry and Vector-Valued Functions 6 hrs

Conic sections, Classifying conic sections by eccentricity, Cylindrical and spherical coordinates, Vector functions, tangent vectors, Curvature and normal vectors, Torsion and binormal vectors.

Unit 6: Multivariable Calculus 7 hrs

Functions of several variables, Limits and continuity, Partial derivatives, Directional derivatives and gradient vectors, Tangent planes, maximum and minimum values, Lagrange multipliers

5. List of Tutorials

The following tutorial activities students shall be conducted to cover all the required contents of this course. This will enable the students to complete the related mathematical problems under the supervision of the subject teacher.

a) Finding limit of a function
b) Finding different asymptotes to a curve
c) Examine the continuity of a function at a point and finding limiting value
d) Find derivative of a function at a point, Show that differentiability implies continuity but the converse may not be true, problems on derivative of implicit function
e) Finding extreme value
f) Problems in Mean value theorems: Rolle's theorem, Lagrange's mean value theorem and related problems
g) Solving problems using L'Hospital's rule
h) Evaluate various indefinite integrals
i) Finding value of Definite integrals, Improper Integrals
j) Evaluation of area. arc length and area of revolution
k) Finding the value using Dot and vector product between 2, 3 and 4 vectors
l) Finding area by scalar product of 2 vectors and volume by scalar product of 3 vectors
m) Finding the projection vector and its value and Reciprocal vector of given 3 vectors
n) Finding equation of line and plane by vector method
o) Classify the conic and finding cylindrical and spherical coordinates
p) Finding tangent vectors, curvature of a given curve and its normal vector
q) Evaluating torsion and binormal vector to given vector
r) Examine the continuity of f(x, y)
s) Finding partial derivative of a function of 2 or 3 variables
t) Finding directional derivative of a vector and finding tangent planes
u) Evaluating extrema using Lagrange multipliers

6. Evaluation Scheme

The evaluation of students' performance in the course will be based on the following components:

Evaluation Method	Description	Weight (%)
Unit Test and Assessment	Regular assignments will be given to assess the understanding and evaluate via internal exam	10 + 30
Tutorial Evaluation	Students will work on individual and submit the assign tutorial	10
Class Participation	Active participation in class discussions and workshops will be taken into account	10
Final examination	Final examination will be conducted to evaluate theoretical knowledge and problem-solving skills	40

7. Recommended Books:

- James Stewart, *Calculus Early Transcendentals*, 7E, CENGAGE Learning.
- Kreyszig, E. *Advanced Engineering Mathematics*. New Delhi: John Wiley and Sons Inc.
- David C. Lay, *Linear Algebra and Its Applications*, 4th Edition, Addison Wesley, Pearson.

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B. Tech. in IT

Course Title: Foundation of Information Technology	**Year: I**	**Credit: 3 (48 hrs)**
Course Code:	**Semester: I**	**Lecture: 2** **Tutorial: 0** **Practical: 1**

1. Course Description

Fundamentals of Information Technology covers the explanation of technology and computer hardware , software ,evolution of computer, peripheral devices and basic network concept. In addition this course also provide concept of emerging technologies related to information technology. This course also provides practical knowledge of MS DOS, Basic Hardware and Networking, Handling GUI based operating system, Office package (Word Processing, Spreadsheet, and Presentation tools) and troubleshooting.

2. Course Objectives

The course aims to achieve the following objectives:

The main objective of this course is to introduce the fundamentals of computing devices and computer system, particularly with respect to personal use of computer hardware and software, the Internet, networking and mobile computing. It focuses on such computer literacy that prepares students for life-long learning of computer concepts and skills.

In addition, this course also makes students capable of using different application packages for their personal as well as professional life.

3. Learning Outcomes

Upon successful completion of the course, students will be able to:

- Understand the fundamental concepts and computer peripheral units.
- Identify the different elements of an information system, including input, output and processing elements.

- Identify the components of a computer system, including hardware components and software applications.
- Describe the purpose and components of database management, knowledge management systems and specialized information systems.
- Identify potential security risks when using an information system, including personal and professional computer systems and networks, and describe forms of computer network and internet security.
- Learning and Explore concept of emerging technologies.
- Learn to use CUI and GUI Interface to operate computer applications.
- Apply different tools and technique to create documents using Word Processing Software
- With summery this subject provides details of major components of the information technology infrastructure of an organization: hardware and software; data resources; telecommunications and networks; and the Internet

4. Lecture Course Details (2 credits/ 32 hrs)

Unit-1 Introduction to Computer System – 3 hrs

Introduction to Computer, Characteristics of Computer, Scope and Applications of Computer, Classification of Computer, Generation of Computer, Anatomy of a Digital Computer, the Computer System

Unit-2 Computer Hardware – 4 hrs

Introduction to hardware, Central Processing Unit, Memory Unit, Instruction Format, Instruction Set, Instruction Cycle, Microprocessor, Interconnecting the Units of a Computer Memory & Its Classification, Input devices, Output Devices

Unit-3 Computer Software - 3 hrs

Introduction to computer software, Types of Software, System Software, Application Software; Utility Software, Operating System: Introduction and Objectives of Operating System, Types of Operating System, Functions of Operating System, Operating System Security, Examples of Operating Systems)

Unit-4 Data Representation 3 hrs.

Number System, Conversion from Decimal to Binary, Octal, Hexadecimal, Conversion of Binary, Octal, Hexadecimal to Decimal, Conversion of Binary to Octal, Hexadecimal, Conversion of Octal, Hexadecimal to Binary, Binary Arithmetic, Signed and Unsigned Numbers, Binary Data Representation, Binary Coding Schemes, Logic Gates

Unit-5 Fundamentals of Database Management System – 3 hrs

Introduction to DBMS, Advantage of Database Management System over Flat file System, Applications of Database Management System, Database Models, Database Language, Data Security, Data Warehouse, Data Mining, Database Users

Unit-6 Data Communication and Computer Network- 5 hrs

Introduction to communication system, Mode of Communication, Introduction to Computer Network, Network Types, Network Topologies, Transmission Media, Network Devices, TCP/IP & OSI Reference Model, Communication Protocols, Centralized vs. Distributed System, Wireless Networking

Unit-7 The Internet and Internet Services – 5 hrs.

Introduction; History of Internet; Internetworking Protocol; the Internet Architecture; Managing the Internet; Connecting to Internet; Internet Connections; Internet Address; Internet Services; Uses of Internet, WWW and Its Evolution, Architecture of Web, Uniform Resource Locator (URL), Browsers and It's Feature, Search Engine, Web Servers: Apache, IIS, Proxy Server; HTTP Protocol, FTP protocol

Unit-8 Computer Security – 2 hrs

Security Threat and Security Attack, Malicious Software, Security Services, Security Mechanisms (Cryptography, Digital Signature, Firewall, Users Identification and Authentication, Intrusion Detection Systems), Cyber Security, Security Management

Unit-9 Emerging Technologies – 4 hrs

Cloud Computing, Mobile Computing, Quantum computing, E-governance, E-commerce, E-Learning, Ubiquitous Learning, E-Banking, Mobile Banking, Crypto Currency, GIS, Artificial Intelligence, Machine Learning, Robotics, Virtual Reality, Deep learning Augmented Reality, Ambient Reality, Internet of Things, Blockchain Technology

5. Tutorial

N/A

6. Practical (1 credit/ 16 hrs)

Part1 - Basic of Computer and Network

- Basic building block of PC, hardware and peripheral introduction
- Assemble and disassemble of computer
- Learning about network cable and crimping different cable types

Part2 – Learn CUI and GUI Based Operating System

- Install different types of operating system
- Learning command line interface such as Command line/terminal
- Learn MS-DOS Internal command and External command

Part 3: Learn word processing software

- Concept of word processing, Introduction of MS Word, Features, Keyboard Shortcuts, Editing & Formatting documents, Table, Advance features of MS Word, Mail Merge, Macros

Part 4: Learn spreadsheets software
Worksheet basic, Working with formulas and cell referencing, Auto sum, Formatting to worksheet, Previewing and Printing worksheet, Graphs and Charts, Functions, Macros

Part 5: Learn presentation software
Introduction, Use of Standard Formatting toolbar, Drawing toolbar, Editing slides, Changing templates, Slide Layouts, inserting clipart & Pictures into slide, Slide Transitions, Animation, Inserting sound and movies into slides

Part 6: Email, Internet and online tools for office management
Learn different email system, configuring email into personal computer, using internet-based tools for office management

7. Evaluation Scheme

The evaluation of student's performance in the course will be based on the following components:

Evaluation Method	Description	Weight (%)
Assignments	Regular assignments will be given to assess the understanding theoretical and practical implementation of basic fundamental computer tools	30
Projects/Seminars	Students will work on individual and group projects to demonstrate their ability to design and prepare office related documents.	20
Class Participation	Active participation in class discussions, labs, and workshops will be taken into account	10
Final examination	Final examination will be conducted to evaluate theoretical knowledge and problem-solving skills	40

8. Recommended Books:

- Goel, A. (2010). Computer Fundamentals. Pearson Education India.
- Education, M. (2002). Peter Norton's Introduction to Computers Fifth Edition, Computing Fundamentals, Student Edition. McGraw-Hill Education.
- Emerging Technologies in Computing(2021): Theory, Practice, and Advances 1st Edition
- Weverka, P. (2019). Office 365 All-in-One for Dummies. John Wiley & Sons.
- Gookin, D. (1998). DOS for Dummies. For Dummies.

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B. Tech. in CS and AI

Course title: C Programming	Year/Part: I/I	Credit 3 (48 hrs)
Course code: **Nature of the Course:** Theory + Lab	Semester: I	Lecture – 2 Tutorial 0 Practical 1

Course Description:

This course combines the principles of computational thinking with hands-on C programming to introduce students to the world of computer science and programming. Students will learn how to approach and solve problems using computational thinking concepts and apply them to implement solutions using the C programming language. Through hands-on programming exercises and projects, students will gain a solid foundation in both computational concepts and C programming.

Course Objectives:

Introduce students to the principles of computational thinking and their relevance in problem-solving. Familiarize students with the basic syntax, data types, and control structures of the C programming language. Develop students' skills in algorithm design, implementation, and analysis.

Foster logical reasoning and systematic problem-solving approaches.

Provide practical experience in developing small-scale C programs to solve real-world problems.

Syllabus:

Unit 1 : Introduction to Computational Thinking (2 Hrs.)

Overview of computational thinking and its significance in problem-solving. Problem-solving strategies and abstraction.

Algorithm design and analysis.

Unit 2 : Introduction to Programming and C Programming (4 Hrs.)

Introduction to programming, types of programming and language translator, algorithm and flowchart; Overview of C and its historical significance, syntax and semantics Installing and setting up the development environment (IDE or text editor).

Unit 3 : Basic Syntax and Data Types (8 Hrs.)

Structure of a C program: header files, main function, and comments. Data types: int, float, char, and their modifiers. C Tokens; C Character Sets; Data types, Constants and Variables; Expression, statements and comments; Escape sequences and Delimiters 1/0 functions; Formatted 1/0 Operators: Arithmetic operators; Relational operators; Logical operators; Assignment operators; Increment and decrement operators; Ternary operator; Bitwise operator; Other operators (comma, sizeof); Expression evaluation; Operator precedence and associativity, Type conversions, Preprocessor directives: include and Macros and their usage.

Unit 4 : Control Statements (8 Hrs.)

Decision-making structures: if, if-else, nested if-else, Switch statement, Looping structures: while, for, do-while. Nesting control structures for complex conditions. Jump statements: break, continue, goto

Unit 5 : Arrays and Strings (5 Hrs.)

Declaring and initializing arrays. Accessing array elements and multi-dimensional arrays. Strings and string manipulation functions.

Unit 6 : Functions (6 Hrs.)

Creating and calling functions. Passing arguments to functions: pass by value and pass by reference. Recursive functions.

Unit 7 : Pointers (4 Hrs.)

Introduction to pointers and memory addresses. Pointer arithmetic and dynamic memory allocation (malloc, calloc, realloc, free). Pointers and arrays.

Unit 8 : Structures and Unions (3 Hrs.)

Defining and using structures. Nested structures and arrays of structures. Understanding unions and their uses.

Unit 9 : File Handling (5 Hrs.)

File handling concepts: opening, reading, writing, and losing files. Sequential and random access file operations. Handling errors during file operations.

Laboratory Works:

Practical lab exercises in the context of C programming language are vital for reinforcing classroom concepts, offering hands-on experience to solidify comprehension. These exercises provide immediate feedback, fostering problem-solving skills, and encouraging student collaboration. Moreover, C programming lab work familiarizes students with industry-relevant tools and libraries, preparing them for real-world programming challenges.

Evaluation Scheme

Evaluation of students' performance is divided into parts: Internal assessment (theory and practical and internal external examinations (theory and practical). The distribution of points is given below:

Internal Assessment		External Assessment		
Theory	**Practical**	**Semester Examination (Theoretical exam)**	**External Practical Exam/ Viva**	**Total Points**
25	15	40	20	100

Note: *Students must pass separately internal assessment, external practical* exam *and semester examination*

Text Books:

Let Us C, Yashavant P. Kanetkar, BPB Publications

Reference Books:

Brian Kernighan and Dennis Ritchie, The C Programming Language, Prentice Hall PTR
Byron Gottfried, Programming with C, McGraw Hill Education